

An Experimental Analysis of Motor Imagery EEG Signals Using Feature Extraction and Classification Methodologies

Praveen Kumar Shukla
Department of Electronics & Telecommunication
National Institute of Technology Raipur
Email: pkshukla.phd2017.elx@nitrr.ac.in

Rahul Kumar Chaurasiya*
Department of Electronics & Telecommunication
National Institute of Technology Raipur
*Corresponding Author: rkchaurasiya@nitrr.ac.in

Abstract- Classifications of motor imagery task is a principal challenge in the brain-computer interface (BCI) application. In this paper, motor imagery of right hand and right foot classification problem has been studied. A discrete wavelet transforms (DWT)-based feature extraction method is proposed. The wavelet-based energy-entropy feature set was used and the quality of features with a variation of scale and wavelet type was examined. Chebyshev filter was used for removal of noise and feature extraction. The EEG data of each channel was divided into windowed frames and DWT is applied on each frame, independently. A set of statistical features were extracted, from the approximate DWT coefficients. The feature set involves wavelet domain energy, entropy, variance, and maximum value of wavelet coefficients. We have used wavelet-based entropy for the five-level decomposition in combination with three different wavelet type viz. Daubechies, Coiflets, and Symlets. This paper aims to identify the best wavelet and decomposition level based on three wavelet type and additionally finds the best classifier perspective of accuracy and receiver operating curve.

Keyword- BCI, Motor imagery, Decision tree, SVM, KNN, DWT.

I. INTRODUCTION

Electroencephalography (EEG) has been adopted by main stream researchers (especially the non-invasive brain computer interface BCI), because of its remarkable temporal resolution, convenience, and low setup costs. The biggest problem in the BCI system is to analyse different EEG signal of distinct motor imagery task in efficient manner. Many researchers have shown their interest in the classification of the motor imagery (MI) task. In literature of MI dataset, preprocessing methods generally involved, 8-30 Hz band pass filter[1], 8-26 Hz band-pass filter[2], 4th order Butterworth IIR Laplacian filter[3], common average reference(CAR) finite impulse response (FIR) filter[4], 8-35 Hz band-pass filter[5], principal component analysis (PCA)[6], Tensor independent component analysis (ICA)[7], and autocorrelation[8]. Whereas for feature extraction, Fast Fourier transform (FFT)(13-16 Hz or 13-30 Hz, 8-12 Hz)[9] (8-30 Hz)[10]continuous wavelet transform (CWT)[11], discrete wavelet transform (DWT) (Daubechies10)[12], and dual tree complex wavelet transform (DTCWT)[13] were used.

Binary classification problem in motor imagery data set of BCI competition II(dataset III) has been generally solved by using LDA[14], HMM[15], and Naive Bayes[16] classifier

for left or right hand classification. Different researchers have used Daubechies wavelet for feature extraction of MI EEG signal. However several other wavelets such as Coiflets, Bi-orthogonal wavelet, Symlets, Morlets and Marxian Hat remains unexplored. In this paper we have explored the possibilities to use Daubechies, Coiflets, and Symlets wavelet for feature extraction purpose. The classification task has been performed using support vector machine (SVM) with versions versions – Linear, Quadratic and Cubic, K-nearest neighbour (KNN) with versions-weighted, fine and medium, and Decision tree classifier with versions- simple, medium and complex tree.

The rest of the paper has been divided into the following sections: section II describes the dataset& method used in this paper, section III shows the experimental result and section IV concludes the paper.

II. MATERIALS& METHODS

A. Dataset

The dataset has been taken from BCI competition III(dataset IVA) in this work[17]. The dataset is for the motor imagery task. According to the data information the data were recorded from 5 healthy people labelled 'aa', 'al', 'av', 'aw' and 'ay'. Subjects were sitting comfortably on a chair with palms resting on armrests. In the data set, signals were collected from 118 electrodes based on 10/20 system. There were 280 cues for 5 subjects, namely 140 trials for every project per subject. The dataset represents that size [time ×channels], mark: structure of target cue information with fields, pos: vector of positions of the cue, y: vector of target classes. During each trial, each subject was required to perform both of the MI task (right hand and right foot) for 3.5 seconds. The data are downsampled were at 100 Hz and sampling rate was 1000Hz. The band-pass filtered between 0.005 Hz and 200 Hz was applied.

B. Methods

This section describes the methodology being used in this paper. The raw EEG signals were first passed through the Chebyshev filter for removing the noise. Further, the discrete wavelet transform was applied to the filtered signals for getting the desired frequency bands. The extracted feature sets were then passed to the classifiers to test the quality of extracted features. The process of preprocessing,

feature extraction, the classifier has been depicted as a flowchart in figure1.

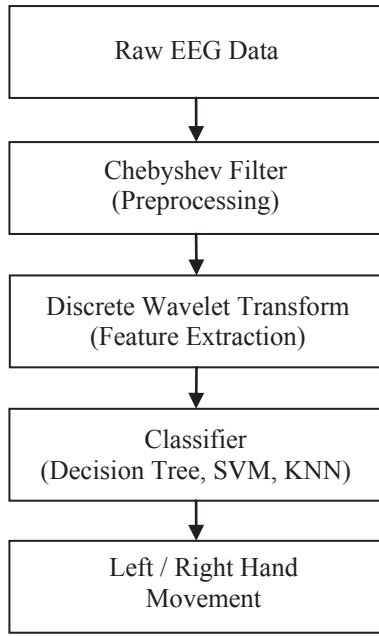


Fig.: 1. Flow chart of motor imagery signal classification.

The detailed description of the aforementioned steps is given in this subsection:

1) *Data Preprocessing*: From the motor imaginary signals a different type of band like Delta (1-4 Hz), Theta (4-7 Hz), Alpha (8-18Hz), Beta (18-25Hz), and Gamma (30-70 Hz) were generated. The raw EEG signals were passed a Chebyshev filter for removing noise as a part of preprocessing of the original data.

2) *Feature Extraction*: In literature, very few feature vectors have been used for left-right imagery challenge discrimination.

We have used DWT-based feature extraction techniques. DWT technique is a multi-resolution technique in time and frequency domain. DWT has the property to have appropriate time resolution at an excessive frequency and good frequency resolution at low frequency. The discrete wavelet coefficient of a signal $X[n]$ is derived as follows in equation (1):

$$x(e, f) = \sum_{m=-\infty}^{\infty} x[n] \phi_{e,f}[m] \quad (1)$$

Where, f is a real number known as window transition parameter, e is called as contraction parameter given by equation (2):

$$\phi_{e,f}[n] = \frac{1}{\sqrt{e}} \cdot \phi \left[\frac{n-f}{e} \right] \quad (2)$$

The condition for dyadic wavelet has been taken as $e = 2^{-j}$, $f = k \cdot 2^{-j}$, so after putting these values in equation (2) we will get our modified result as given in equation (3):

$$\phi_{e,f}[m] = 2^{j/2} \cdot \phi[2^j \cdot n - k] \quad (3)$$

Wavelet filter: DWT is performed by passing signal $x[n]$ first through a low pass filter and then through high pass filter to break the signal into the scale. The output coefficient by the low pass filter or scaling function output is shown in equation (4):

$$\phi(t) = 2 \cdot \sum_{q=0}^m h(q) \cdot \phi(2t - q) \quad (4)$$

The output of the high pass filter is given by equation (5):

$$W(t) = 2 \cdot \sum_{q=0}^m g(q) \cdot \phi(2t - q) \quad (5)$$

The approximate coefficient is further divided into new approximate and detail coefficient. This decomposition process is repeated until the required frequency response is achieved from the input signal. As we have applied different wavelet transforms for feature extraction, a description of different wavelet families has been given in Table - I.

TABLE I WAVELET FAMILIES

S N	Wavelet	Description
1	Daubechies	Daubechies family denoted as signed dbN (N is the order) this wavelet belong to the orthogonal wavelet.
2	Coiflets	A coif lets wavelet denoted as "coif". This wavelet function has 2N movement equal to 0 and the scaling function has 2N-1 movement equal to zero. The function has supported of length 6N-1.
3	Biorthogoal Wavelet	This biorthogonal family wavelet is signed as "bior" biorthogonal wavelet exhibit the perpendicular of linear phase which is needed signal and image reconstruction. Wavelet is divided into two component one for decomposition other for the reconstruction.
4	Symlets	The symlet wavelet is dented "symN" (N is the order). The symlets are nearly symmetrical orthogonal and biorthogonal.
5	Morlets	This wavelet has no scaling function.
6	Marxian Hat	This wavelet has no scaling function derived from a function that is praposnal to the second derivative function of the Gaussian portability density function.

The following four features were extracted from wavelets Daubechies, Coiflets and Symlets.

(i) *The maximum value of wavelet coefficient*: The maximum frequency can be described as in equation(6),

$$\text{Max (frequency band 1, frequency band 2, ...)} \quad (6)$$

(ii) *The variance of the wavelet coefficient*: The variance can be found through the following equation (7),

$$\text{Variance (frequency band 1, frequency band2, ...)} \quad (7)$$

(iii) *Wavelet domain energy*: The wavelet domain energy is computed through the Parseval's theorem described in equation(8)

$$\int x(t).y(t)dt = \frac{1}{2\pi} \int X^*(w).Y(w)dw \quad (8)$$

(iv) *Wavelet domain entropy*: Fundamentals of information theory are applicable to wavelet-based entropy. For a discrete variable X having (N+1) number of possible values ranging between $x_0, x_1, x_2, \dots, x_n$ the Entropy is defined as in equation(9),

$$H(X) = -\sum_{i=1}^n p(x_i) \log_2(P(x_i)) \quad (9)$$

Where equation P (x_i) means the likelihood event for a specific value x.

3) Classification of motor imagery tasks:

The classification was done using SVM, KNN, and Decision tree classifiers (with their different versions). This subsection describes about classifier used to classify the motor imagery tasks into right hand and right foot class.

(i) *Support Vector Machine*: SVM is large margin classifier with better generalization ability & less over-fitting. We have used three versions viz. linear SVM, Quadratic SVM, and cubic SVM (Manual kernel scale =9.8)in this paper[18, 19].

(ii) *K-Nearest Neighbour*: KNN is a lazy learner where the limit is simply approximated locally and all estimations are surrendered until the point when the classification is finished. We are using three versions viz. fine KNN, weighted KNN and medium KNN. K= 10 value is used in this paper for classifying motor imagery task[19].

(iii) *Decision Tree*: A decision tree is a multistage decision system in which classes are successively dismissed until the point that we achieve an at last acknowledged. We are using three versions viz. simple tree, medium tree, and complex tree in this paper for classifying motor imagery task[19].

C. Performance Criterion:

Single performance criterion may not justify a classifier performance completely. If we increase the number of the performance criterion, then the result will be more reliable. Hence this paper uses two performance criterion viz. accuracy and ROC.

1) *Receiver Operating Curve (ROC)*: In a binary classifier, ROC is commonly used to visualize the performance of the predictor. ROC is also called as an AUC (area under the curve).AUC is the best to way to see the performance of the predictor in a single value. ROC curve measure true positive value in Y-axis and false positive value in the x-axis[20].

AUC is computed by adding a point (FP/N, TP/P) every time through moving the threshold value 0 to1. The optimal threshold value is that value which gives the best specificity and best sensitivity value of the model. The ROC area chart and specification for the prediction are taken from R. Chatterjee, et al [21].

2) *Accuracy*: Accuracy is another performance criterion to measure the quality of the binary classifier. Accuracy is defined as the proportion of the correct outcome means both true positive and true negative among the total instance.

III. EXPERIMENTAL RESULTS

We have analysed 1st level to 5th level decomposition to Daubechies, Coiflets, and Symlets wavelets for feature extraction, and classified the signals using Quadratic SVM, Weighted KNN and Complex decision tree. These versions of different classifiers were selected as they outperformed other versions of that classifier (as in Table V). The results thus obtained using different wavelets (with 1st to 5th level decomposition) are shown in Table-II.

TABLE II RESULT AT DECOMPOSITION LEVEL 5 USING DAUBECHIES

Wavelet Order	Performance Matrix	Classification Method		
		SVM	Decision Tree	KNN
Db1	Accuracy	65.9	73.8	99.7
	ROC	0.61	0.85	0.99
Db2	Accuracy	56.5	73.5	98.7
	ROC	0.55	0.84	0.99
Db3	Accuracy	57.6	74.5	98.9
	ROC	0.54	0.81	0.99
Db4	Accuracy	57.4	83.8	99.0
	ROC	0.46	0.82	1.00
Db5	Accuracy	64.5	94.3	95.0
	ROC	0.62	0.92	1.00
Average Accuracy		59.93	79.9	98.3
Average Roc		0.55	0.85	0.99

In this paper, we have examined the accuracy for all five subjects 'aa', 'al', 'av', 'aw', and 'ay'. We have used 10-fold cross-validation technique for training and testing. The binary classification of EEG data into right hand and right foot classes has been achieved through DWT-based features. On comparison of results shown in Table II, III & IV, it can be observed that Daubechies wavelet with db-5 (i.e. 5th level decomposition) has performed better than other methods. Hence we have used the feature extracted from db-5 wavelet to compose the performance of different versions of all three classifiers (i.e. simple tree, medium tree, and complex tree, weighted KNN, fine KNN, medium KNN, linear SVM, quadratic SVM and cubic SVM). The results obtained by classifier are shown in Table –V.

TABLE III RESULT AT DECOMPOSITION LEVEL 5 USING COIFLETS

Wavelet Order	Performance Matrix	Classification Method		
		SVM	Decision Tree	KNN
Coif1	Accuracy	59.7	74.5	98.7
	ROC	0.58	0.82	1.0
Coif2	Accuracy	53.7	74.7	98.7
	ROC	0.51	0.84	1.00
Coif3	Accuracy	56.1	75.0	97.9
	ROC	0.83	0.54	0.99
Coif4	Accuracy	58.3	73.0	95.7
	ROC	0.51	0.83	0.99
Coif5	Accuracy	62.3	76.0	95.6
	ROC	0.57	0.81	0.99
Average Accuracy		58.0	74.6	97.3
Average Roc		0.60	0.68	0.99

TABLE IV RESULT AT DECOMPOSITION LEVEL 5 USING SYMLETS

Wavelet order	Performance matrix	Classification method		
		SVM	Decision tree	KNN
Sym1	Accuracy	63.0	75.0	99.0
	ROC	0.57	0.84	0.99
Sym2	Accuracy	59.7	75.4	97.6
	ROC	0.59	0.87	1.00
Sym3	Accuracy	53.3	73.5	98.4
	ROC	0.53	0.85	0.99
Sym4	Accuracy	57.3	73.8	97.8
	ROC	0.55	0.83	1.00
Sym5	Accuracy	60.4	75.8	97.7
	ROC	0.59	0.84	1.00
Average accuracy		58.75	74.7	98.1
Average ROC		0.566	0.846	0.99

TABLE V CLASSIFICATION ACCURACY OF THE SUBJECT

Method	Classification Accuracy					
	aa	al	av	aw	ay	Average
Simple Tree	95.9	80.4	95.7	70.9	73.8	83.34
Medium Tree	98.7	87.4	99.5	79.9	98.7	92.84
Complex Tree	98.7	88.5	99.5	85.5	98.7	94.18
Weighted KNN	100	91.1	99.3	85.1	99.0	94.90
Fine KNN	100	90.4	99.4	85.5	98.8	94.82
Medium KNN	100	87.4	99.0	81.3	98.4	93.20
Linear SVM	62.1	49.9	57.2	56.1	65.9	58.24
Quadratic SVM	100	58.5	52.6	57.6	53.6	64.46
Cubic SVM	100	50.6	52.4	49.4	51.9	60.86

IV. CONCLUSION

The aim of this paper was to find the best classifier and validate the best wavelet among Daubechies, Coiflets&Symlets, and find the best level of decomposition. Daubechies wavelet with order 5 (based on average accuracy) gives the best result than all other wavelet. This paper validates that KNN is best suitable classifier based on the performance criterion with highest accuracy of 98.3% and ROC approximately tending to 1. Through the work was limited to only three wavelets, in future, we will include more features and investigate with more wavelet type to extend the work.

REFERENCES

- [1] J. Li and L. Zhang, "Active training paradigm for motor imagery BCI," *Experimental brain research*, vol. 219, pp. 245-254, 2012.
- [2] L. Yao, *et al.*, "A novel calibration and task guidance framework for motor imagery BCI via a tendon vibration induced sensation with kinesthesia illusion," *Journal of neural engineering*, vol. 12, p. 016005, 2014.
- [3] W. Song, *et al.*, "Mobile robot control by BCI based on motor imagery," in *Intelligent Human-Machine Systems and Cybernetics (IHMSC), 2014 Sixth International Conference on*, 2014, pp. 383-387.
- [4] Y. Wu and Y. Ge, "A novel method for motor imagery EEG adaptive classification based biomimetic pattern recognition," *Neurocomputing*, vol. 116, pp. 280-290, 2013.
- [5] B. Zhou, *et al.*, "Robust spatial filters on three-class motor imagery EEG data using independent component analysis," *Journal of Biosciences and Medicines*, vol. 2, p. 43, 2014.
- [6] R. Chaurasiya, *et al.*, "Statistical wavelet features, PCA, and SVM based approach for EEG signals classification," *International Journal of Electrical, Computer, Electronics and Communication Engineering*, vol. 9, pp. 182-186, 2015.
- [7] N. Sharma and J.-C. Baron, "Does motor imagery share neural networks with executed movement: a multivariate fMRI analysis," *Frontiers in human neuroscience*, vol. 7, p. 564, 2013.
- [8] S. K. Bashar, *et al.*, "Identification of motor imagery movements from eeg signals using dual tree complex wavelet transform," in *Advances in Computing, Communications and Informatics (ICACCI), 2015 International Conference on*, 2015, pp. 290-296.
- [9] T. Hiroyasu, *et al.*, "Electroencephalographic method using fast Fourier transform overlap processing for recognition of right-or left-handed elbow flexion motor imagery," in *Computational Intelligence in Brain Computer Interfaces (CIBCI), 2014 IEEE Symposium on*, 2014, pp. 24-29.
- [10] J. Jin, *et al.*, "Recognition of right and left motor imagery based on energy features," *JOURNAL-EAST CHINA UNIVERSITY OF SCIENCE AND TECHNOLOGY*, vol. 33, p. 536, 2007.
- [11] W.-Y. Hsu and Y.-N. Sun, "EEG-based motor imagery analysis using weighted wavelet transform features," *Journal of neuroscience methods*, vol. 176, pp. 310-318, 2009.
- [12] B. Xu and A. Song, "Pattern recognition of motor imagery EEG using wavelet transform," *Journal of Biomedical Science and Engineering*, vol. 1, p. 64, 2008.
- [13] L. Wang, *et al.*, "Application of Hilbert-Huang transform for the study of motor imagery tasks," in *Engineering in Medicine and Biology Society, 2008. EMBS 2008. 30th Annual International Conference of the IEEE*, 2008, pp. 3848-3851.
- [14] C.-Y. Chen, *et al.*, "A novel classification method for motor imagery based on Brain-Computer Interface," in *Neural Networks (IJCNN), 2014 International Joint Conference on*, 2014, pp. 4099-4102.
- [15] B. Obermaier, *et al.*, "Hidden Markov models for online classification of single trial EEG data," *Pattern Recognition Letters*, vol. 22, pp. 1299-1309, 2001.
- [16] H. Wang and Y. Zhang, "Detection of motor imagery EEG signals employing Naïve Bayes based learning process," *Measurement*, vol. 86, pp. 148-158, 2016.
- [17] S. Imran, *et al.*, "Motor imagery EEG signal classification scheme based on wavelet domain statistical features," in *Electrical Engineering and Information & Communication Technology (ICEEICT), 2014 International Conference on*, 2014, pp. 1-4.
- [18] R. K. Chaurasiya, *et al.*, "An efficient P300 speller system for Brain-Computer Interface," in *Signal Processing, Computing and Control (ISPCC), 2015 International Conference on*, 2015, pp. 57-62.
- [19] P.-N. Tan, *et al.*, *Introduction to Data Mining. (First Edition)*: Addison-Wesley Longman Publishing Co., Inc., 2005.
- [20] R. Chatterjee, *et al.*, "Effects of wavelets on quality of features in motor-imagery EEG signal classification," in *Wireless Communications, Signal Processing and Networking (WiSPNET), International Conference on*, 2016, pp. 1346-1350.
- [21] R. Chatterjee and T. Bandyopadhyay, "EEG based Motor Imagery Classification using SVM and MLP," in *Computational Intelligence and Networks (CINE), 2016 2nd International Conference on*, 2016, pp. 84-89.